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SAMPLE DOCUMENT USING `ADENC.STY`

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Check out the Github Repository for `adenc.sty`.

1. Package Options

The following package options are supported:

- `color` adds background colors for theorem environments (see Section 2).
- `plain` uses the default theorem environments (`definition`, `plain`, and `remark`); `boxed` adds a box around the theorem environments. You may choose between one of the two (default is `plain`).
- `hideproofs` and `hidemarkings` hide, respectively, proof environments and markings generated using the `\markabove` and `\markbelow` commands (see Section 3).
- `workingpaper` adds (1) a watermark with date on the first page to indicate that the current document is a draft and (2) more space to the margin so that notes written with the `\todo` command (using the `todonotes` package) can fully display.

To pass an option, use: `\usepackage[Option]{adenc}`.
To pass multiple options, use: `\usepackage[Option1, Option2, ...]{adenc}`.

2. Theorem Environments

Definition 2.1. A definitive **definition** is a definition, by definition.

Lemma 2.2. A lamentable lemma.

Theorem 2.3. A towering theorem.

Corollary 2.4. A cool corollary.

Remark 2.5. A remarkable remark.

Example 2.6. An exemplary example.

Problem 2.7. A problematic problem.

Proof. A precise proof. □

Proposition 2.8 (A long proposition that breaks across pages). Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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Numbering can be turned off by using the corresponding `*` versions of the environments (e.g. `theorem*` instead of `theorem`).

3. Marking the Document

The `\todo` command in the `todonotes` package is a great way to add notes to a document, but among other things, it does not support display style math and, when used frequently, the places to which they point can be hard to decipher. It is for these reasons that the following commands are introduced:

E.g., this points to the word "can."

And this to "hard."

(a) `\begin{itodo} ... \end{itodo}` (inline todo) produces an inline block of notes. This can be used as placeholders for contents to be added later.

This is an example of what notes produced by the `itodo` environment looks like. Unlike the `\todo` command, the `itodo` environment supports display math:

$$\sum_{n=1}^{\infty} a_n z^n.$$

(b) `\markabove` and `\markbelow` provide a way to mark texts without altering the spacing. Both commands take two arguments: (1) align method (`l`, `c`, or `r`); and (2) text to display. For example,

Test test test test test
I'm marking below here
Test test test test.
above!
Test test test test
math! α

is produced by the following code:

```
Test test test\markabove{l}{test1} test\markbelow{c}{I'm marking below here}.

Test test test test.

Test test test\markabove{c}{above!} test\markbelow{r}{math! \(\alpha\)}.
```

4. New Commands

Some commands (mainly for math symbols) are added or modified for aesthetics and/or convenience. A few notable ones are mentioned below:

Description	**Example**	**LaTeX Commands**
Command for styling new vocabulary[1]	**vocab**	`\vocab{vocab}`
Contradiction symbol	$\divideontimes$	`\contradiction`
Shortcuts for `\mathbb`	$\mathbb{R}, \mathbb{Q}, \mathbb{F}, \mathbb{P}$	`\RR, \QQ, \FF, \PP`
Shortcuts for `\mathcal`	$\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$	`\cA, \cB, \cC, \cD`
Shortcuts for `\mathsf`	$\mathscr{L}, \mathscr{T}, \mathscr{U}, \mathscr{V}$	`\sL, \sT, \sU, \sV`
Better looking complement symbol	A^{c}	`A^\complement`
Better looking empty set symbol	$\varnothing$	`\emptyset`
Command for vectors	$\mathbf{v}$	`\vec{v}`
Shortcuts for matrices	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	`\bmat{1 & 2 \\ 3 & 4}`
	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	`\pmat{1 & 2 \\ 3 & 4}`
	$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$	`\vmat{1 & 2 \\ 3 & 4}`
Differentiation operators	$\mathrm{d}x, \mathrm{D}f$	`\d x, \D f`
(Use `\dd` in integrals for correct spacing.)	$\int f \, \mathrm{d}x$	`\int f \dd x`
Imaginary number	i	`\I`
The indicator function	$\mathbb{1}$	`\ind`
Independent	$\perp\!\!\!\perp$	`\indep`

[1] In, for example, definitions.

5. Credits

I have stolen a lot of stuff from below:

- https://web.stanford.edu/~lindrew/lindrew.sty
- https://github.com/gillescastel/lecture-notes
- https://tex.stackexchange.com/questions/142242/robust-way-to-mark-draft-text
- https://math.stackexchange.com/questions/160039/are-there-any-symbols-for-contradictions